IDS Projects

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2024-2025

Schedule

S11	10/03	6	Choice of project + P2P	1.5		
S12	17/03	7	P2P + clouds	1.5	Project design brainstorming & start	1.5
S13	24/03	8	Meeting with TEACHERS		Project - 1st simple version	3
S14	31/03	9			Project	1.5
S15	07/04	10	Revision : exam from last year	1.5	//// deadline 21/04	
S16	14/04	11	Defenses		Defenses	6
S17	21/04	Spring VACATION	////			

Administrativia

- Organisation
  - 2-3 members per group
  - *people that turn in alone will get the grade 0*
- Project advancement:
  - Discussions per group, design principles, report, etc.
- Turn in your project
  - use a git repository for your project (private)
  - turn in report on Moodle
  - demo session - April 14th, the whole day

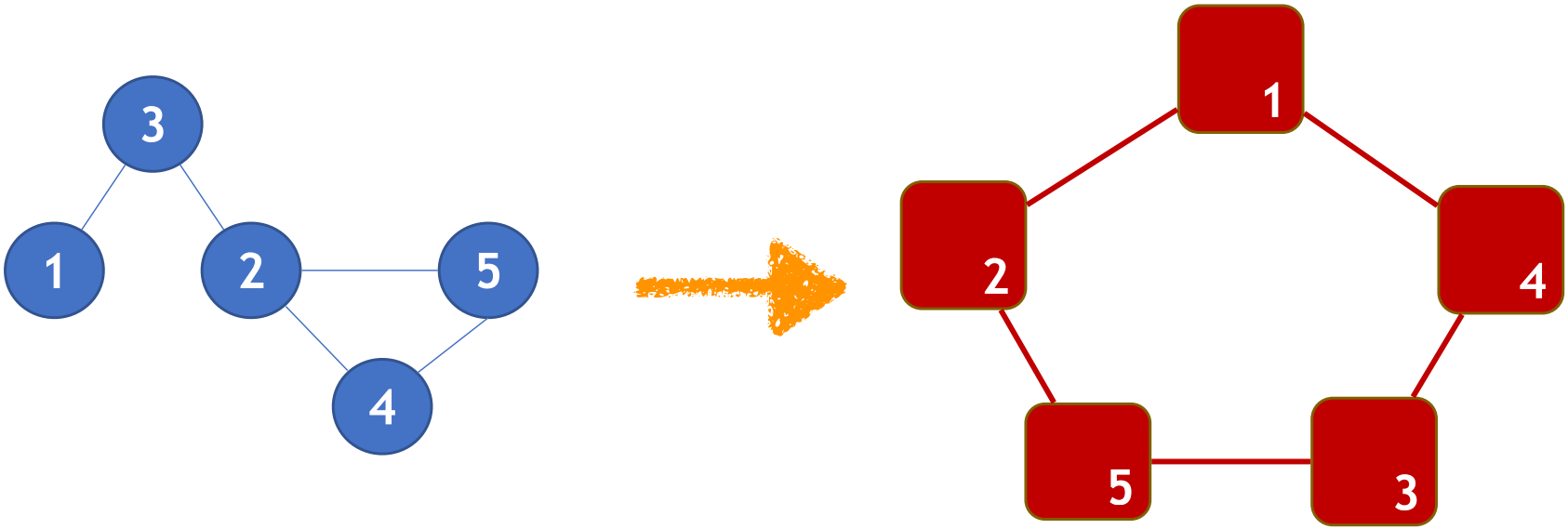
The project

1. Work on the algorithmic logic
   - write the algorithm with rules or describe a protocol / sequence of messages
   - and be sure to cover all the cases
2. Implement with the techno (RMI or RabbitMQ) you want
   - usually RabbitMQ is the smart choice
   - or use both and compare the solutions en terms of software engineering, design simplicity, performance ...
3. Explain the choices (algo/techno) you made
   - the demo is very short, be sure to mention the major ideas
   - be sure to present your choices in the report
4. Make it work in a distributed way
   - the different nodes are launched as different and independent JVM
   - need to bootstrap the system

Project 1 : Overlay

1. The goal is to construct a virtual network over a physical one

2. We start with a graph that gives the physical topology and build a virtual topology

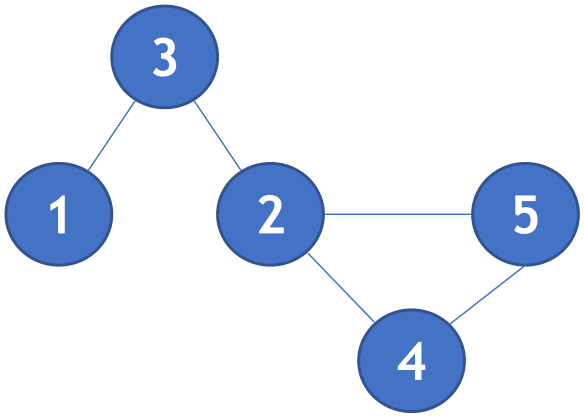


Overlay : the input graph

1. Manage an input describing a bi-directionnal graph e.g an adjacency matrix
   - be able to work with different (types of) graphs
2. The graph gives the physical topology of your system
   - you need to instantiate it with RabbitMQ or Java RMI to have a running distributed system

	1	2	3	4	5
1			1		
2			1	1	1
3	1	1			
4		1			
5		1			

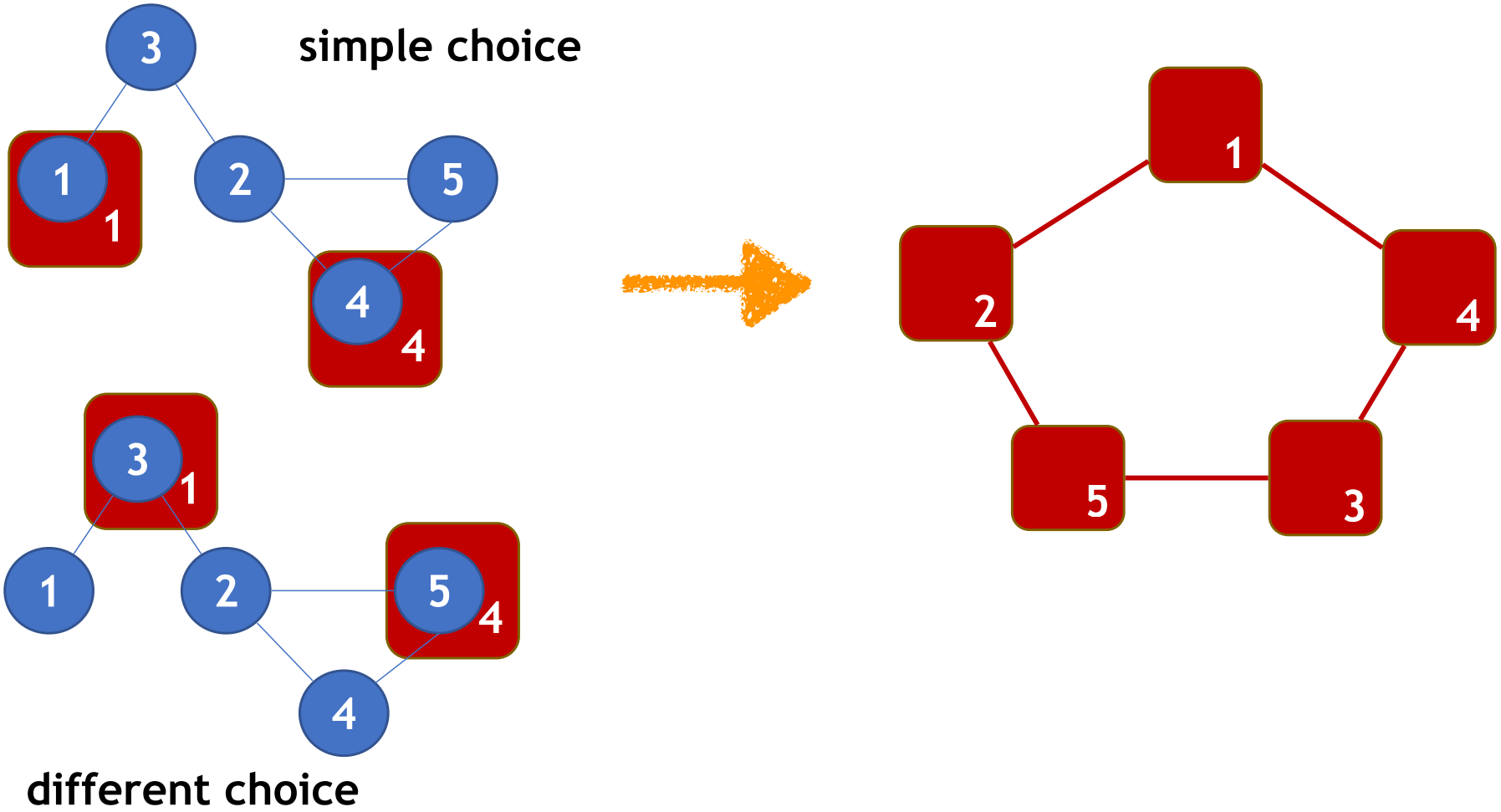
input matrix



running system, each node is a separate running entity (e.g. a remote object in RMI)

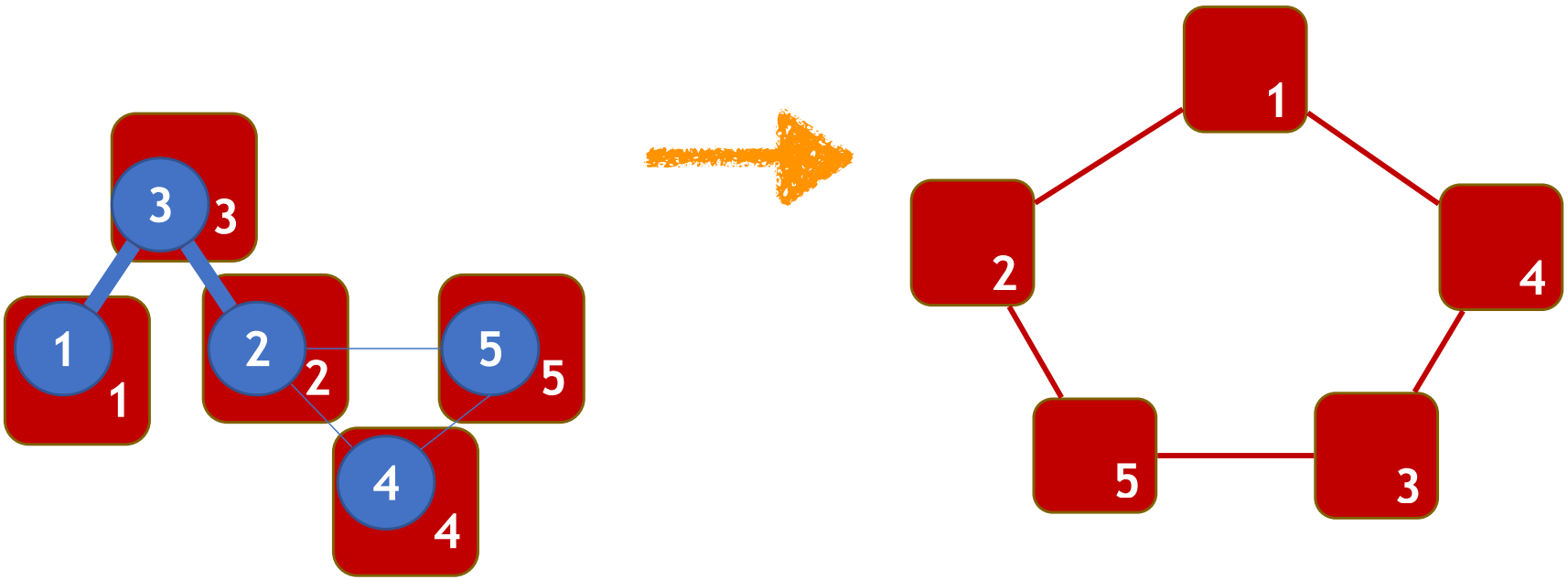
Overlay : virtual ring

- Decide of the nodes' identifiers



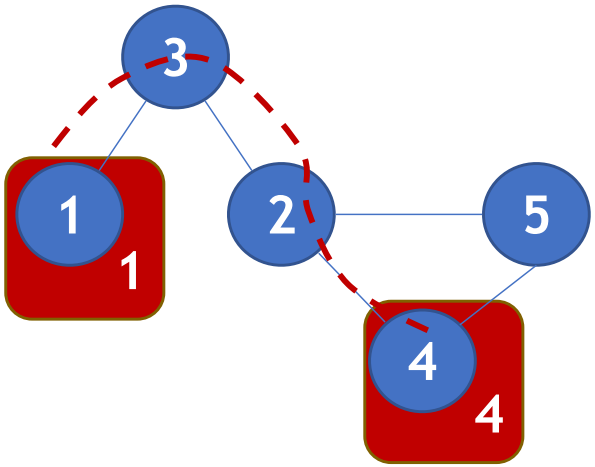
Overlay : virtual ring

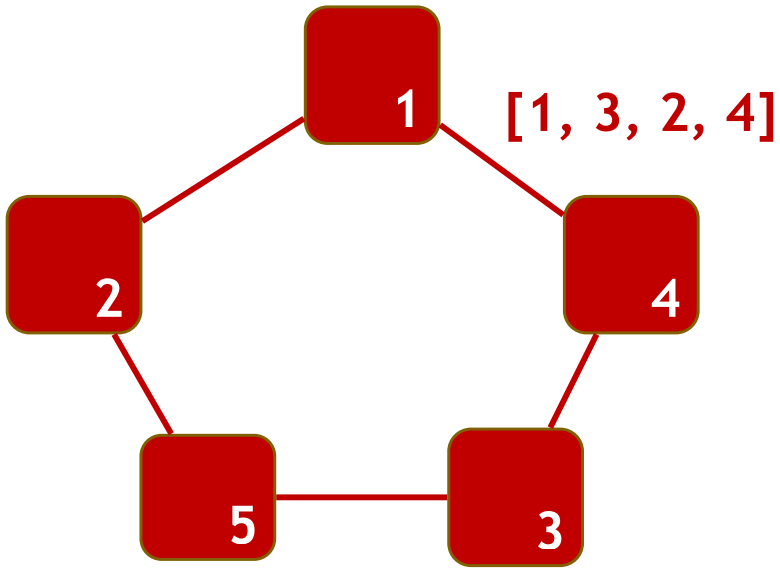
- Decide who is neighbor of who in the virtual ring
  - May be 1-2-3-4-5-1 or, as given in the picture, 1-4-3-5-2-1



Overlay : virtual ring

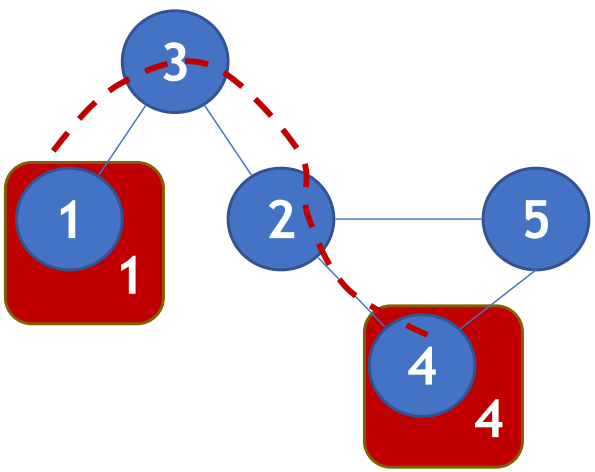
- Compute the routing between a ring node and its two neighbors

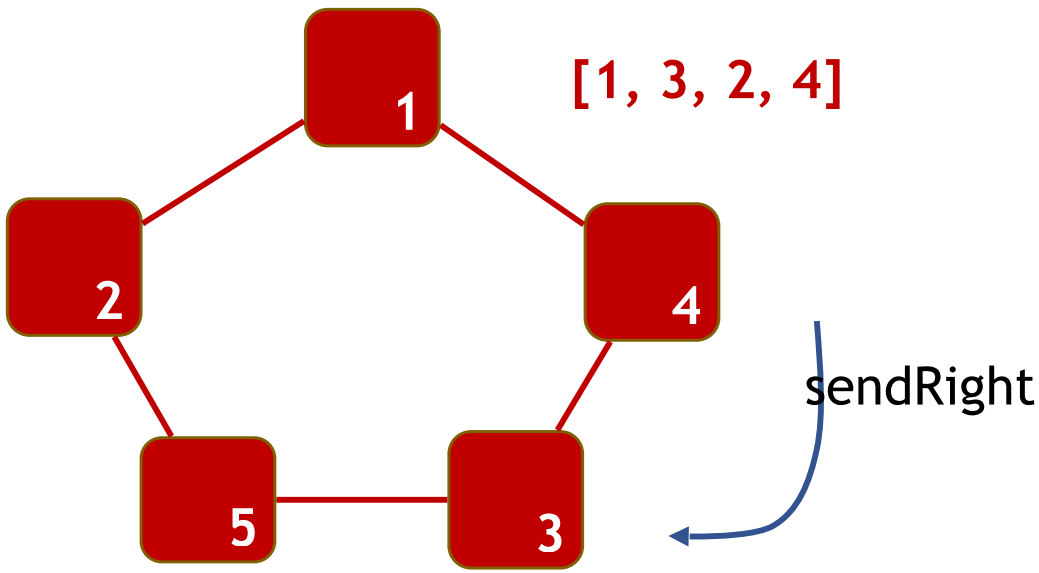




Overlay : virtual ring

- Implement the two basic primitives
  sendLeft(Message m) and sendRight(Message m)



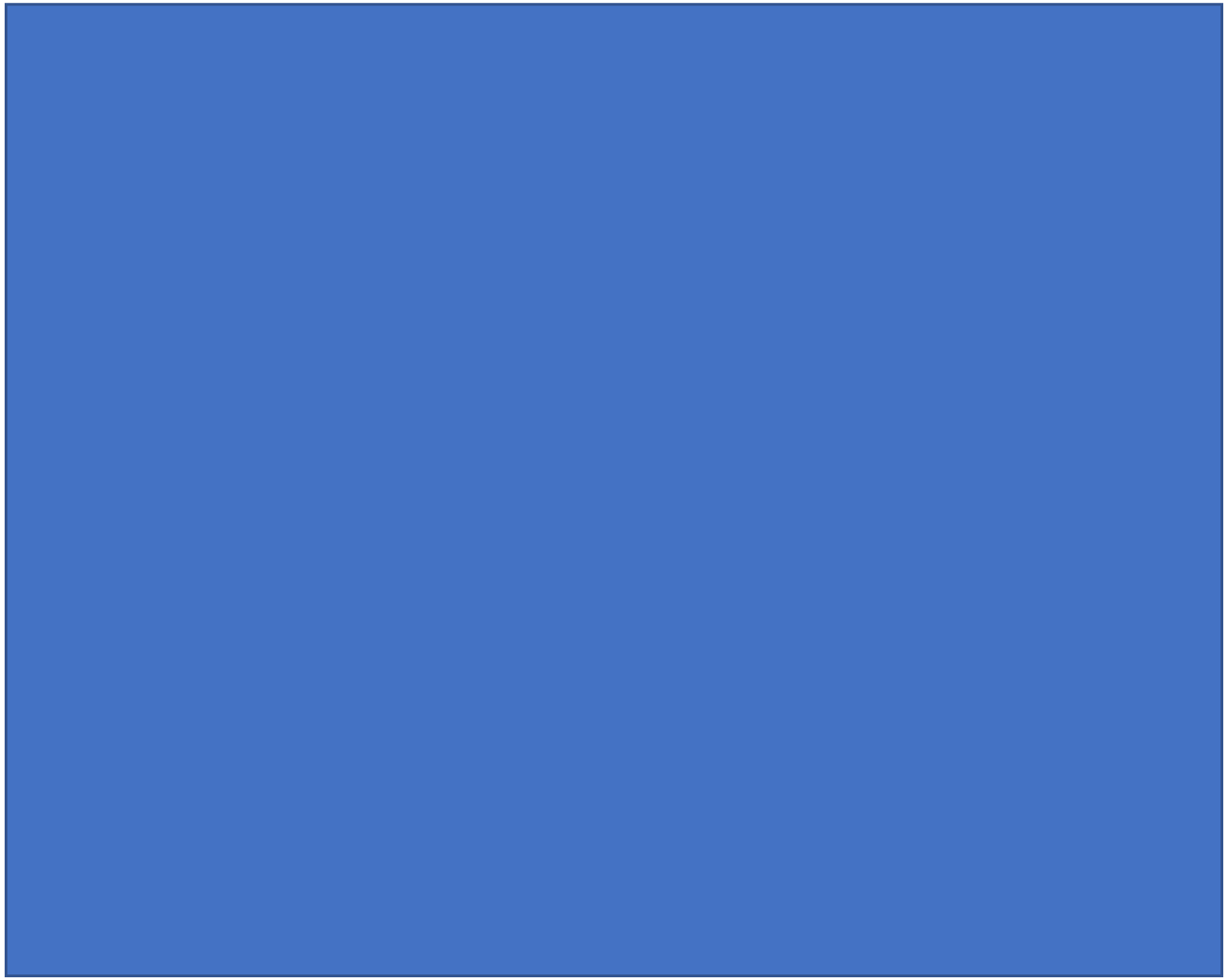


Project 2: Multi-player Game

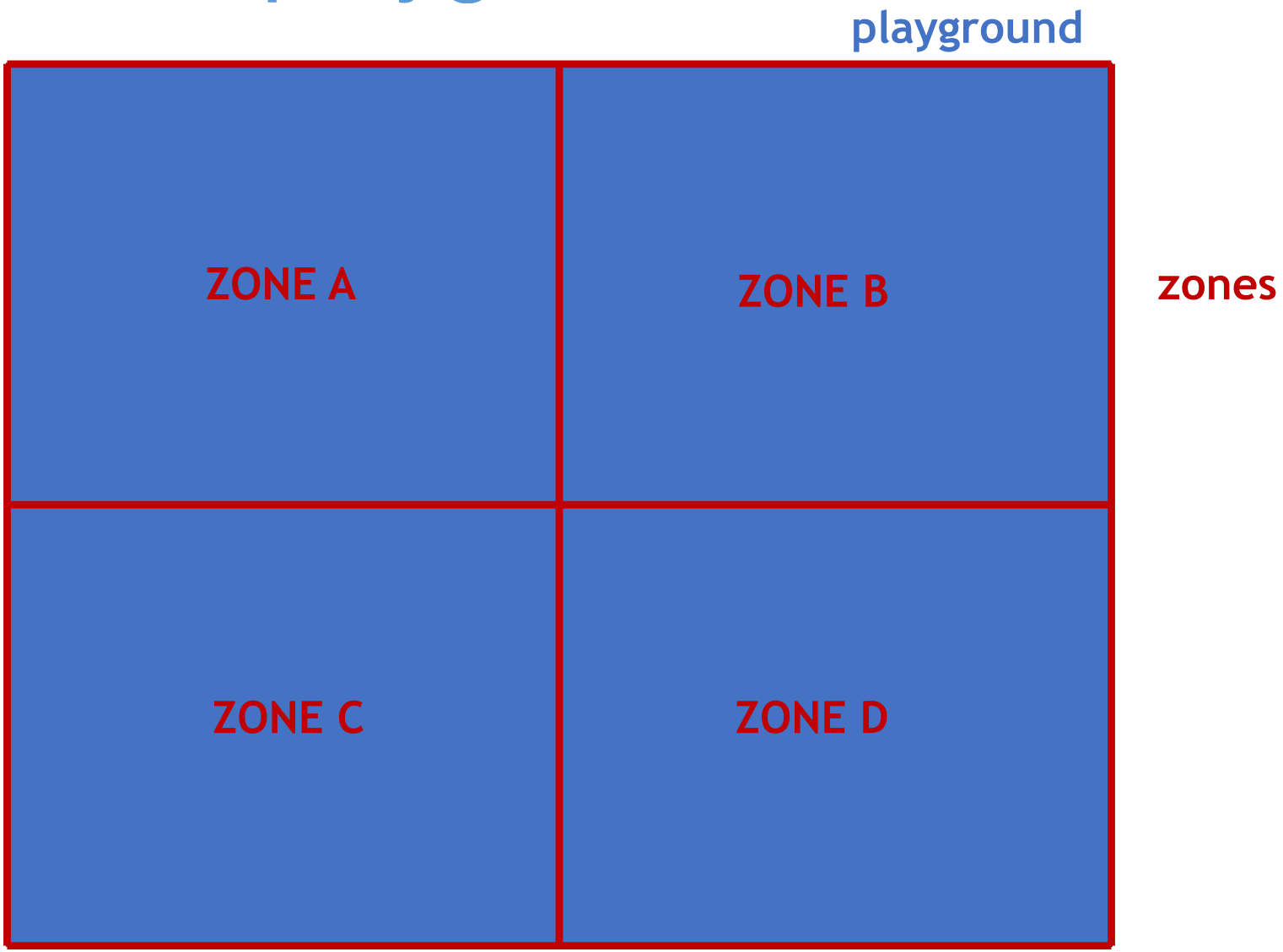
- Goal : Follow the movements of players
- Setting
  - The playground is separated in zones
    - you need to decide how
  - Players move around and pass from one zone to another
    - how do they move?
    - should avoid collisions
    - should say "Hello" to each other
  - Each zone is managed by a node
  - The nodes that manage the set of zones are connected in a distributed topology
    - You need to choose what topology

Project 2: Multi-player Game (2)
EXAMPLE of a setting

playground



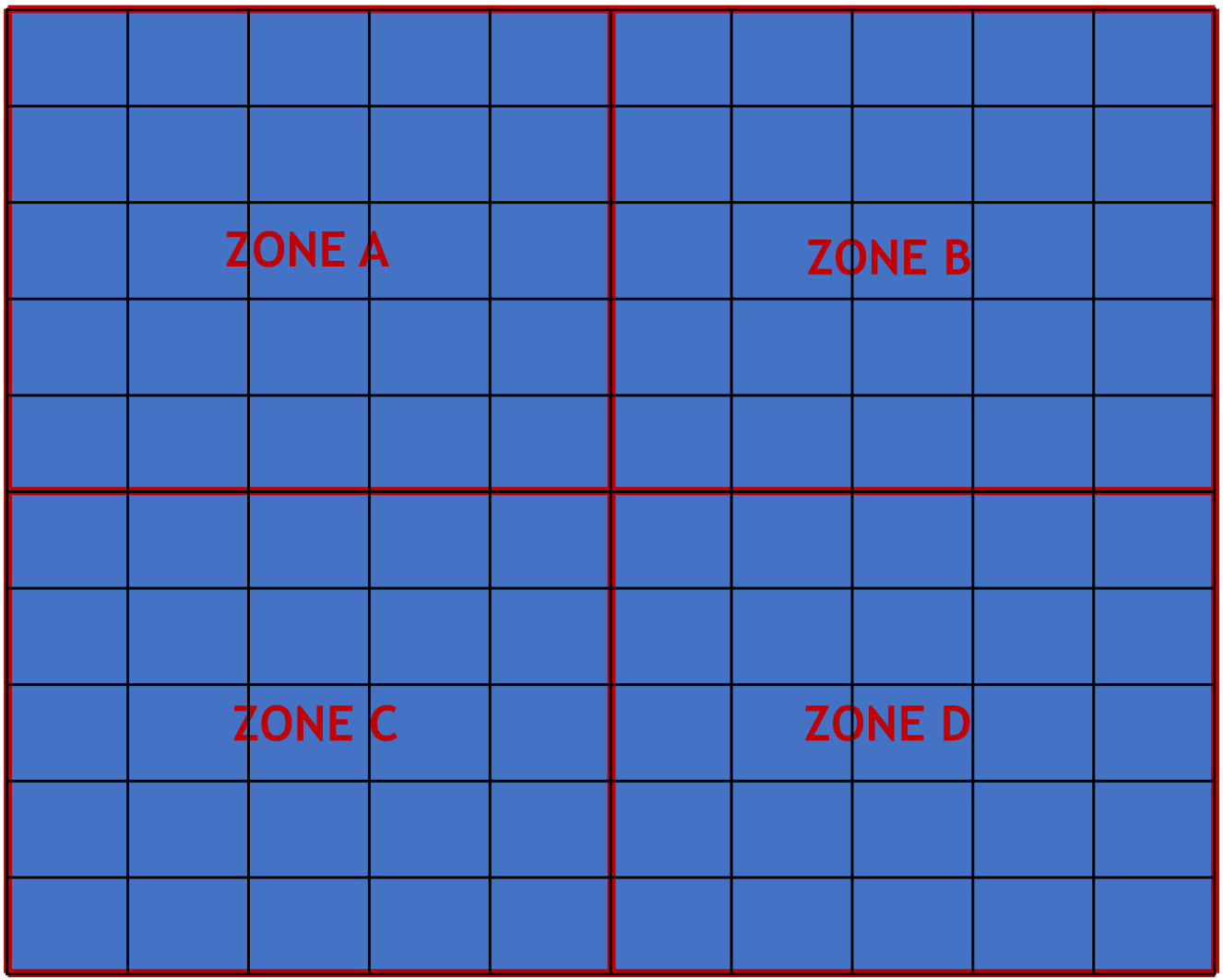
Project 2: Multi-player Game (2)
Divide the playground



Project 2: Multi-player Game (2)
Possible player positions

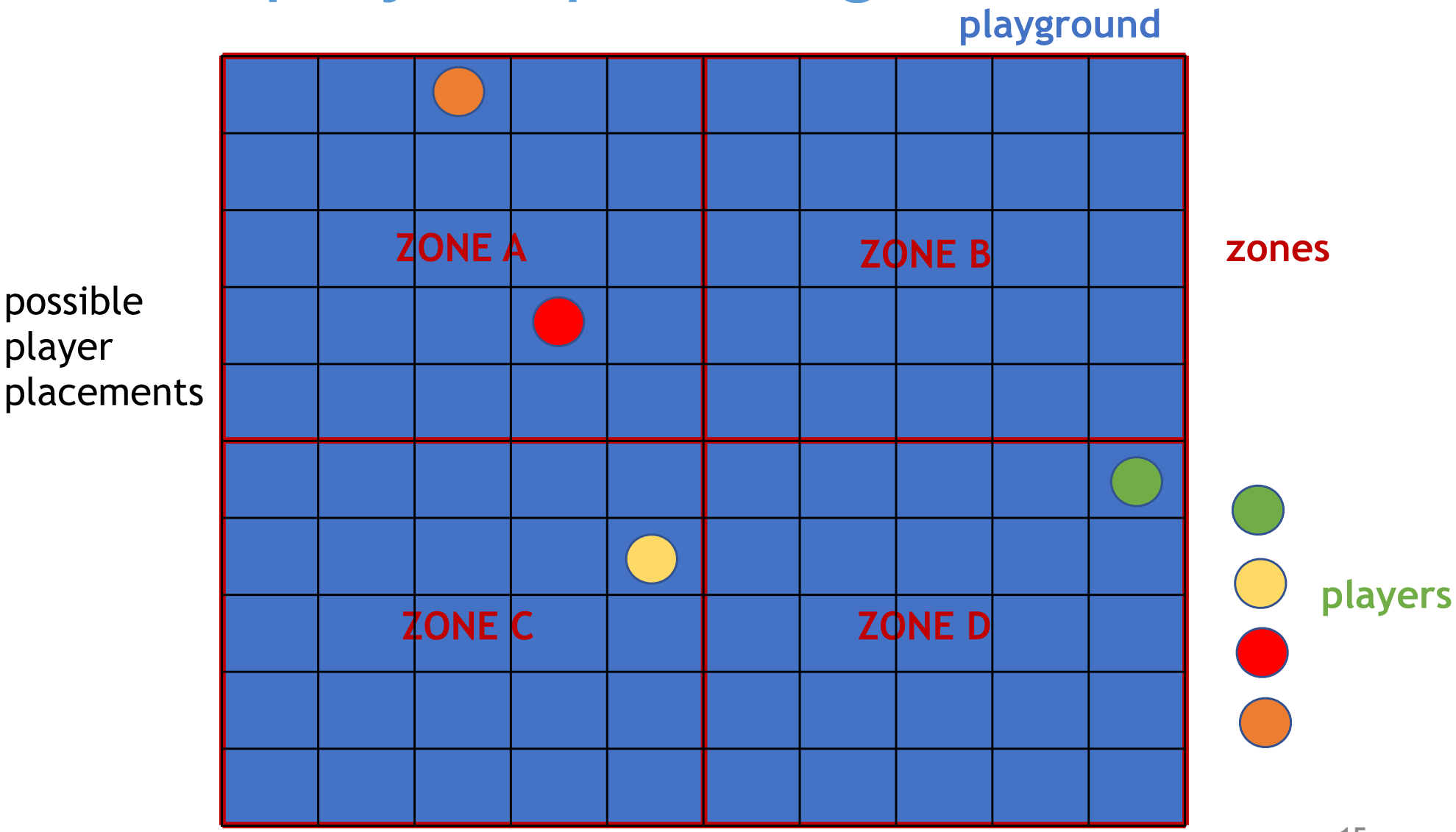
playground

possible
player
placements

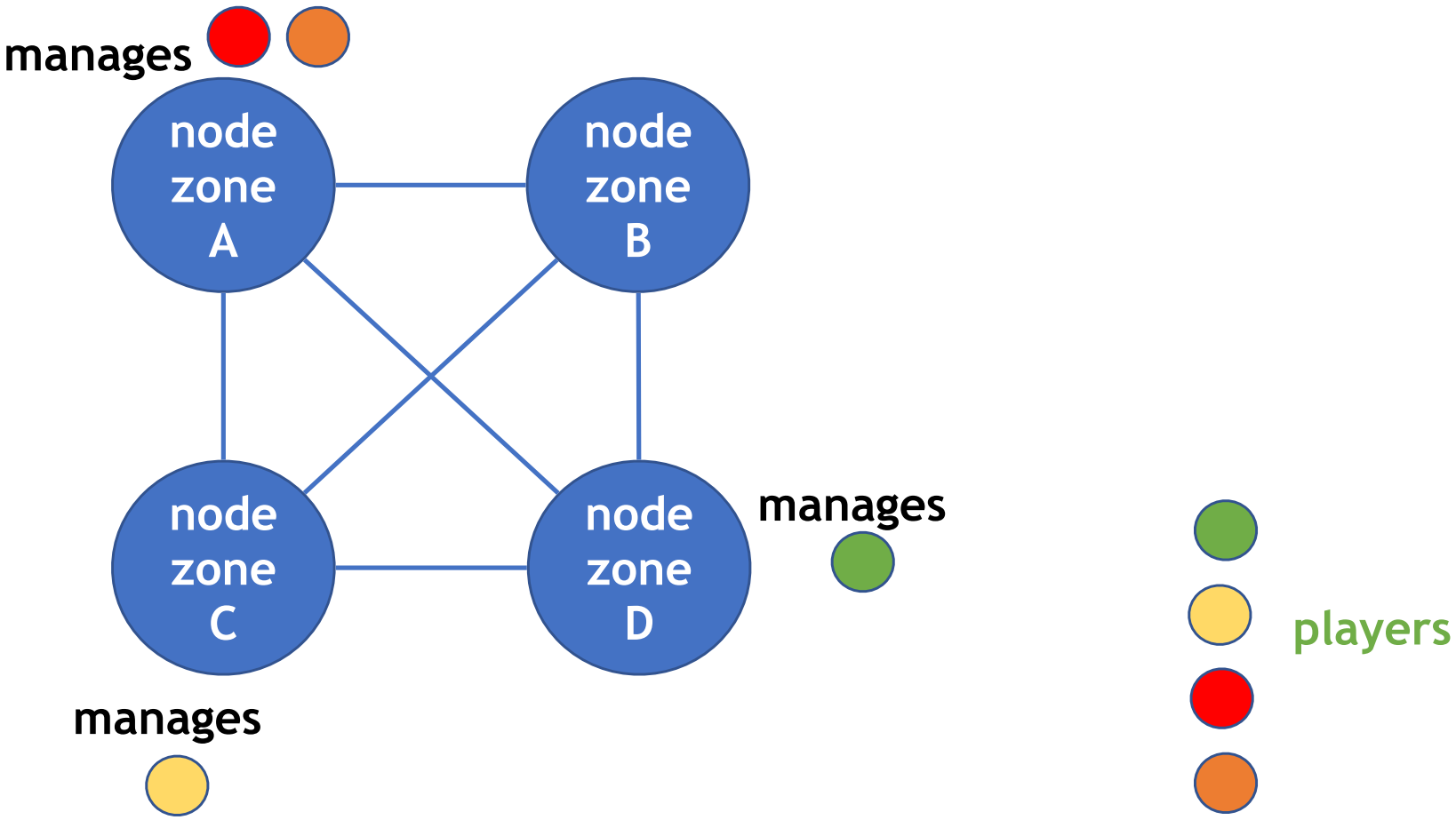


zones

Project 2: Multi-player Game (2)
Initial player spawning

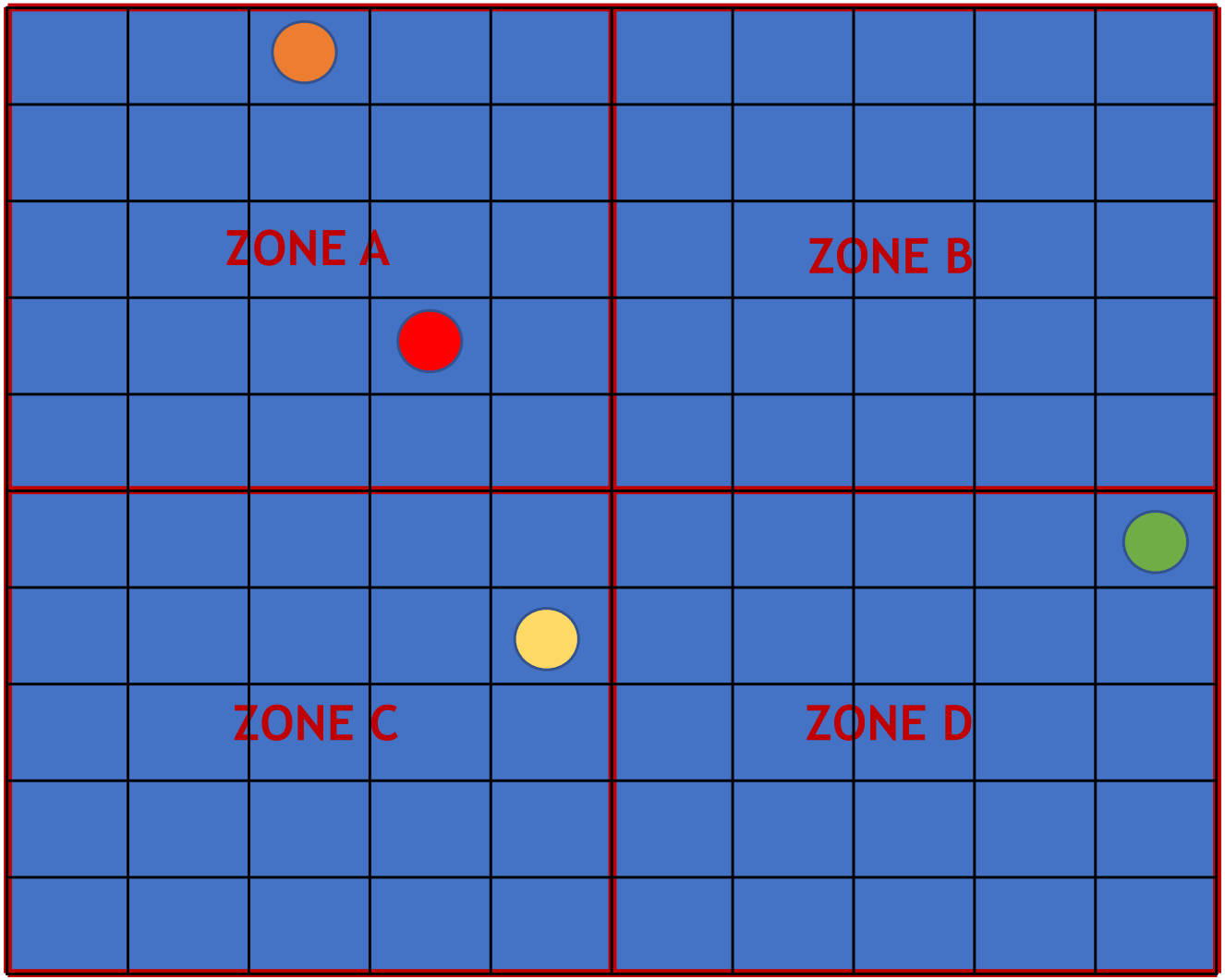


Project 2: Multi-player Game (2)
Node interconnexion



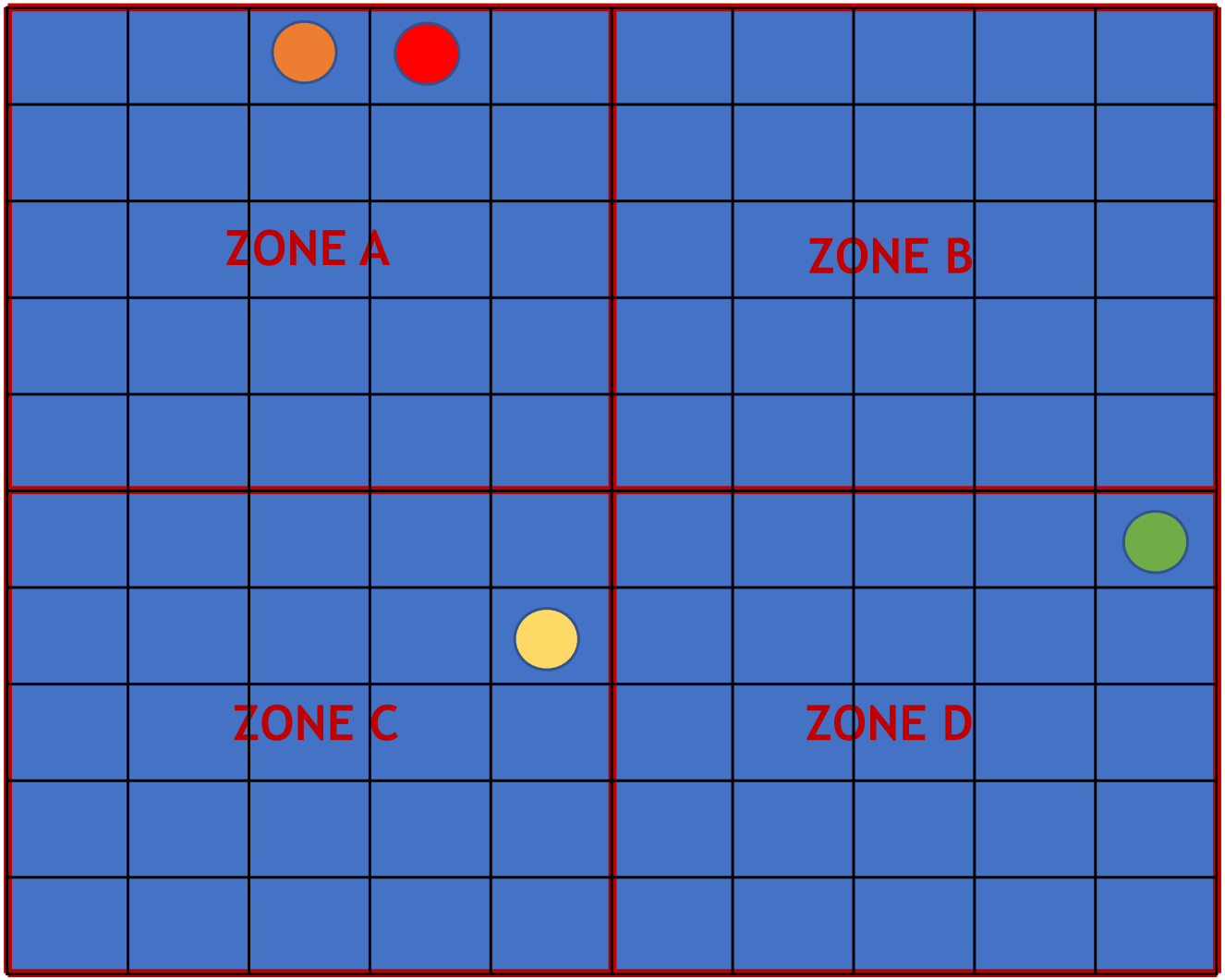
Project 2 : Multi-player Game (2)

Red will move...



Project 2 : Multi-player Game (2)

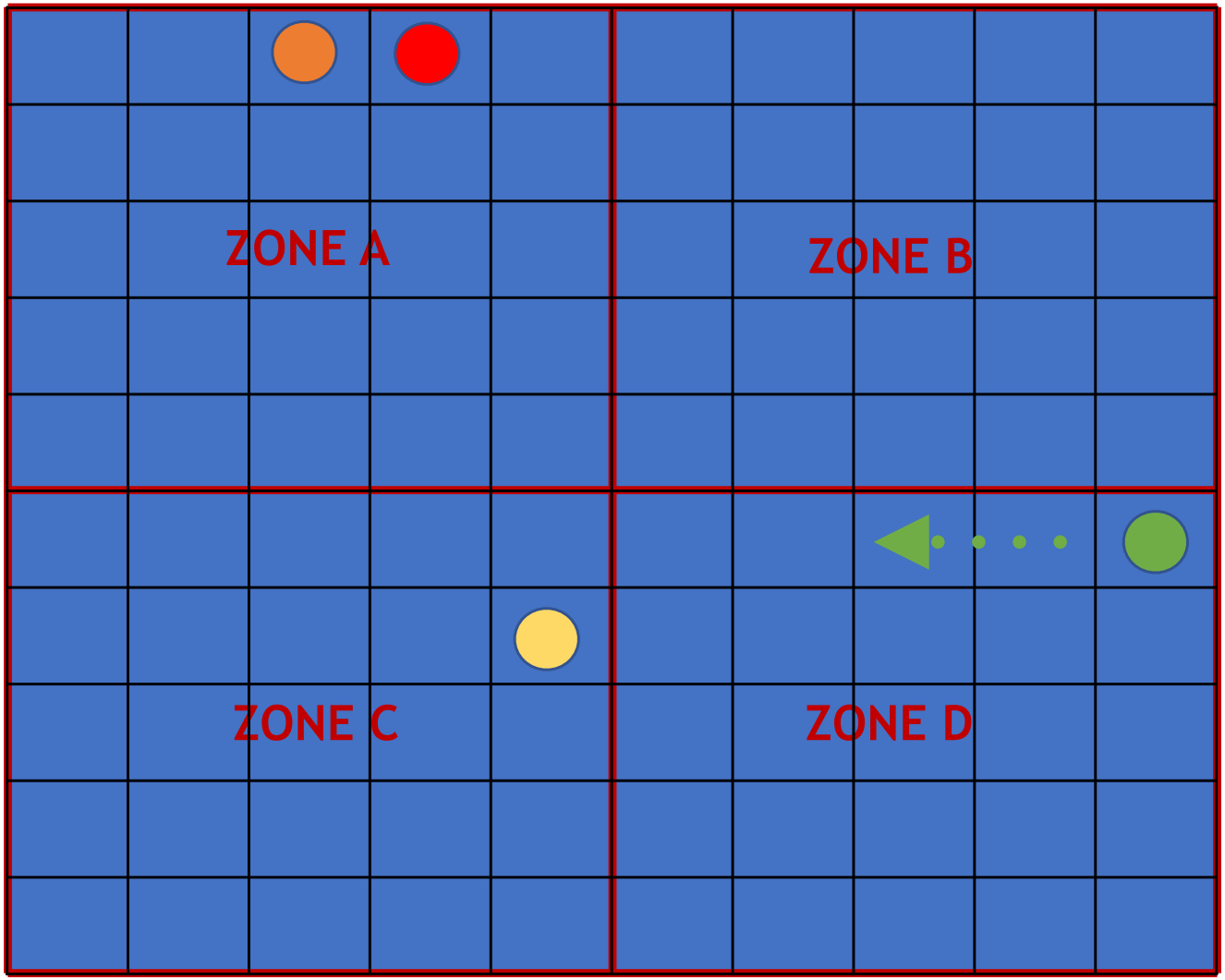
Red says Hello to Orange



Green enters zone C

Project 2 : Multi-player Game (2)

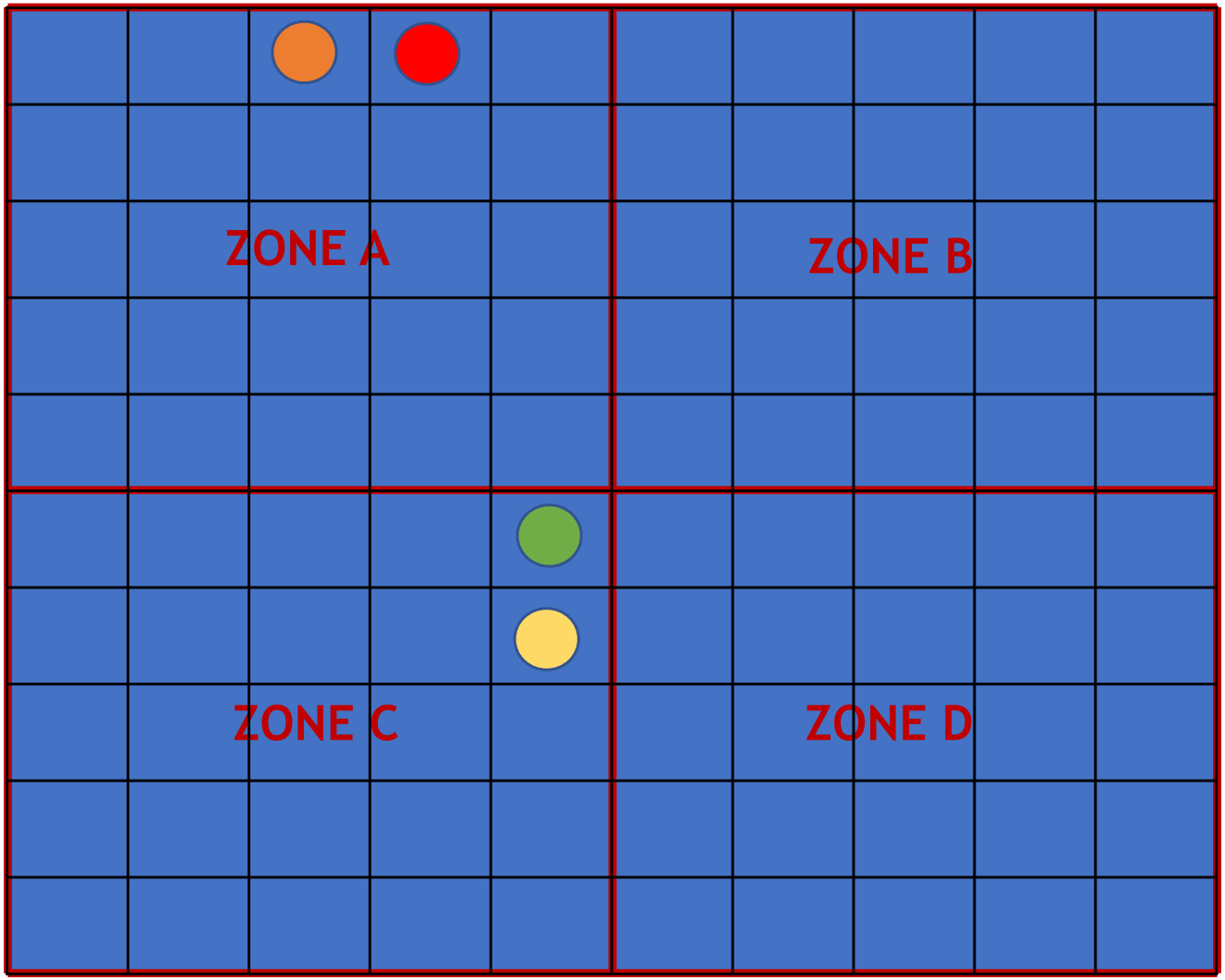
Red says Hello to Orange



Green moves...

Project 2 : Multi-player Game (2)

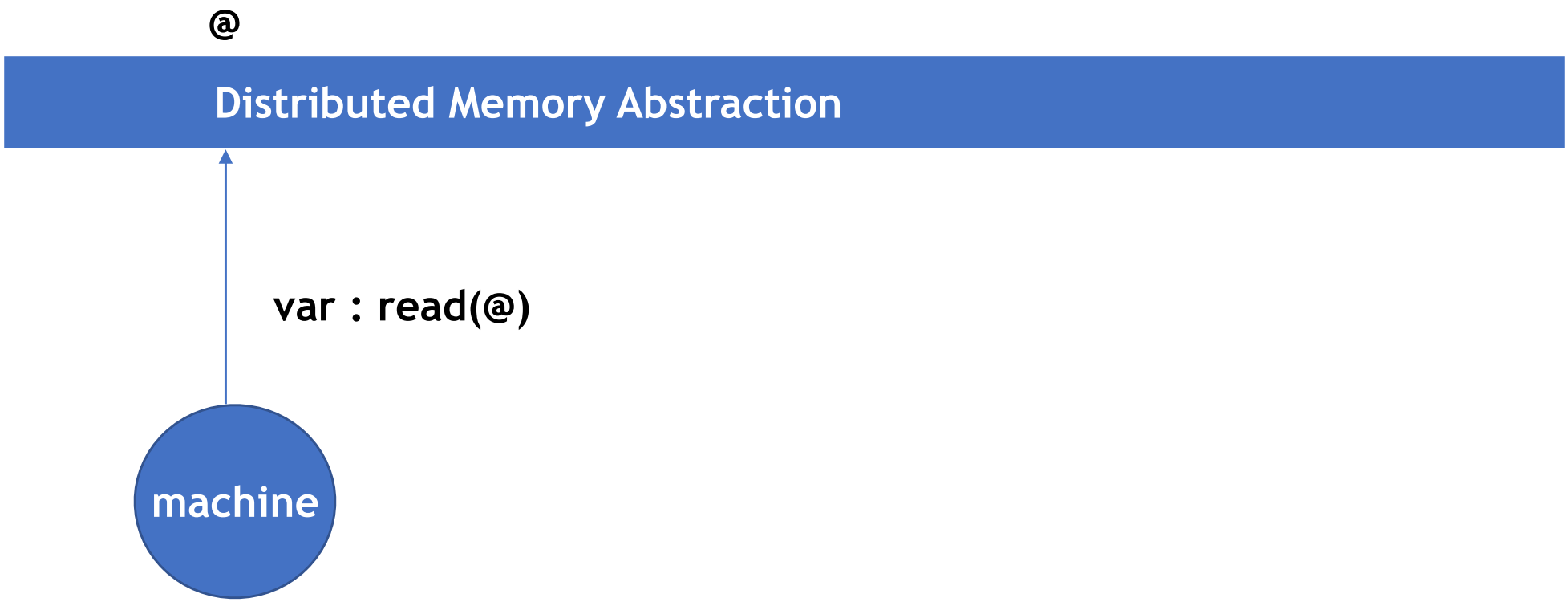
Red says Hello to Orange



Green enters zone C, and says Hello to Yellow

Project 3: DSM (Distributed Shared Memory)

- Distributed Shared Memory Abstraction



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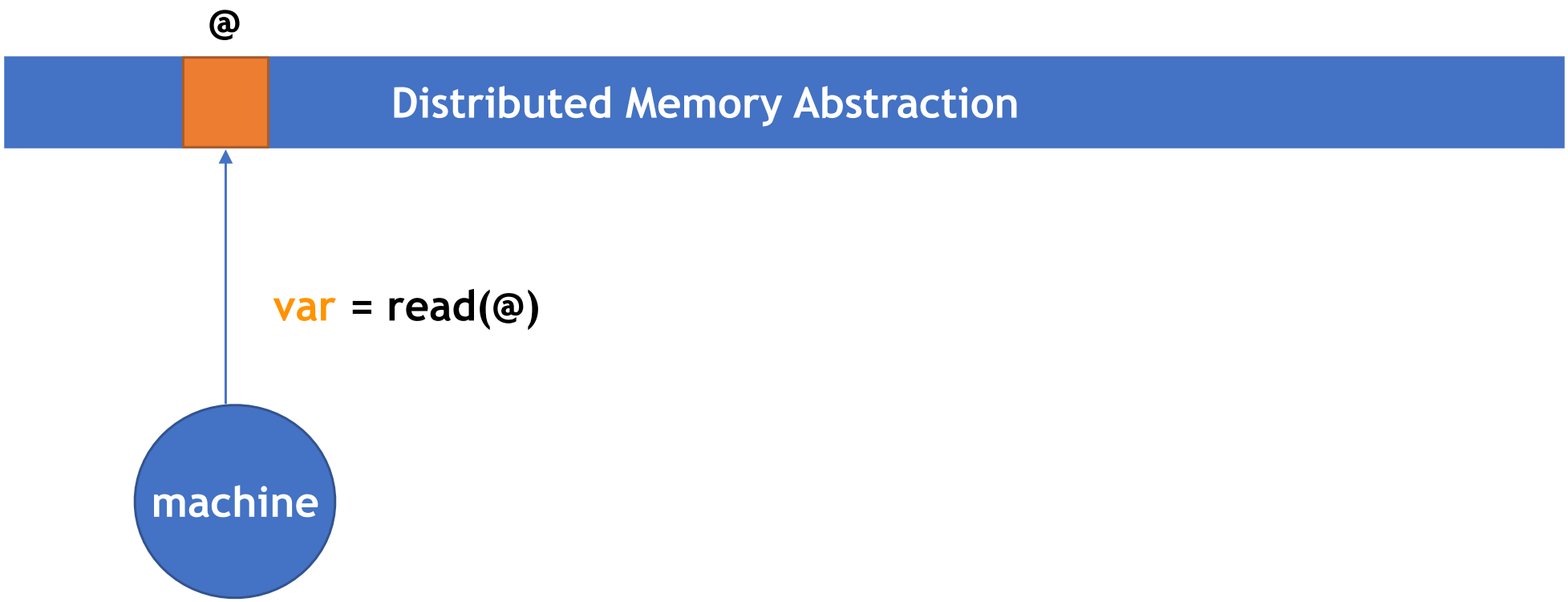

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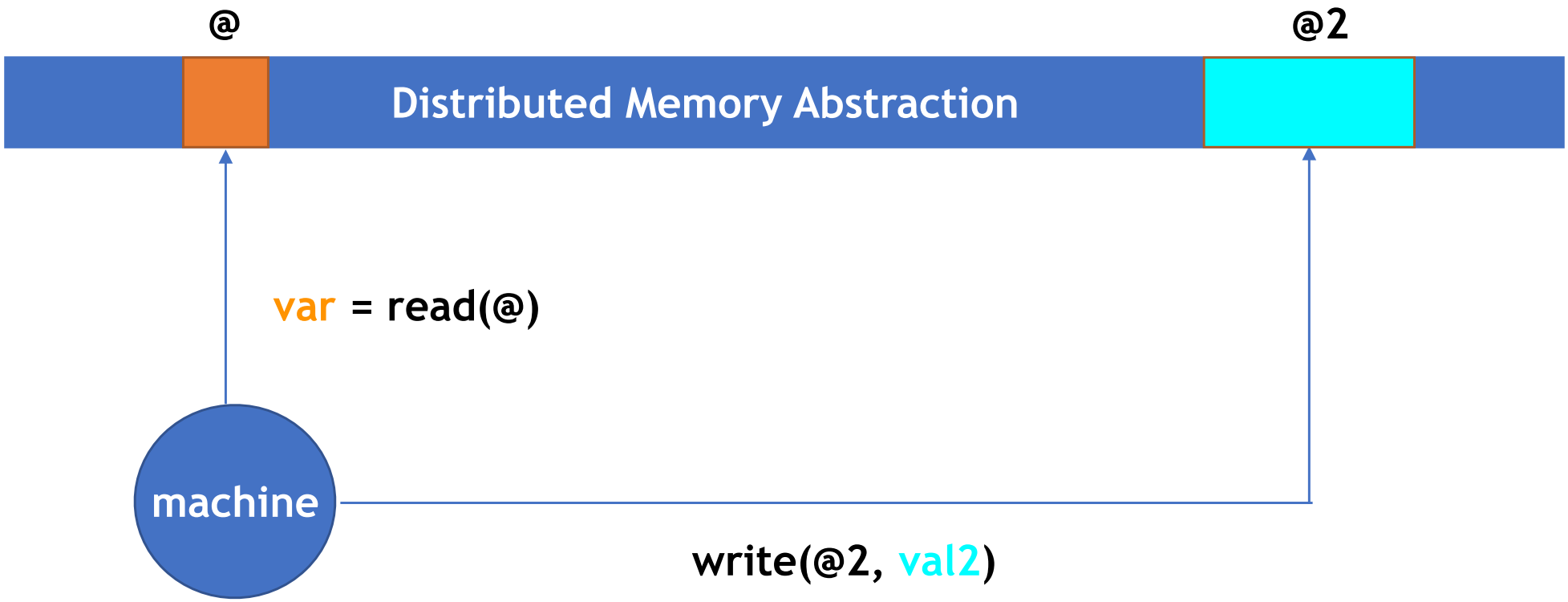


Project 3: DSM (Distributed Shared Memory)

Interface

```
• value read(address)
• write(address, value)
```

1. You need to decide what is an address and how to manage addresses
2. You need to decide what kind of values you will manipulate

```
• value read(address, size)
• write(address, size, value)
```

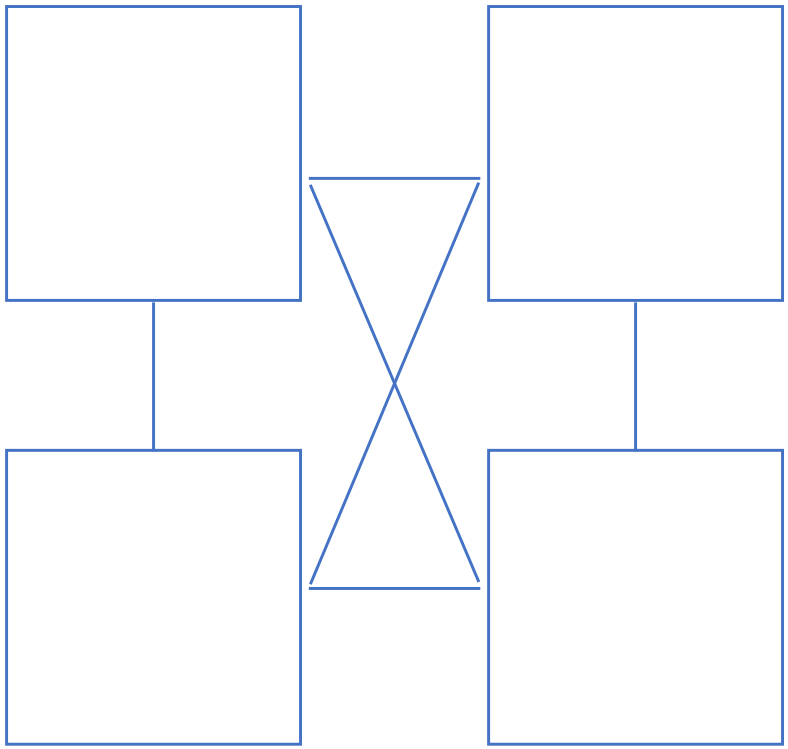
Project 3: DSM (Distributed Shared Memory)

- The memory is distributed over several physical machines
  - You need to decide how to distribute the memory
- The machines are interconnected into a distributed topology
  - You need to choose the topology

- You need to decide what is an address and how to manage addresses
- You need to decide what kind of values you will manipulate

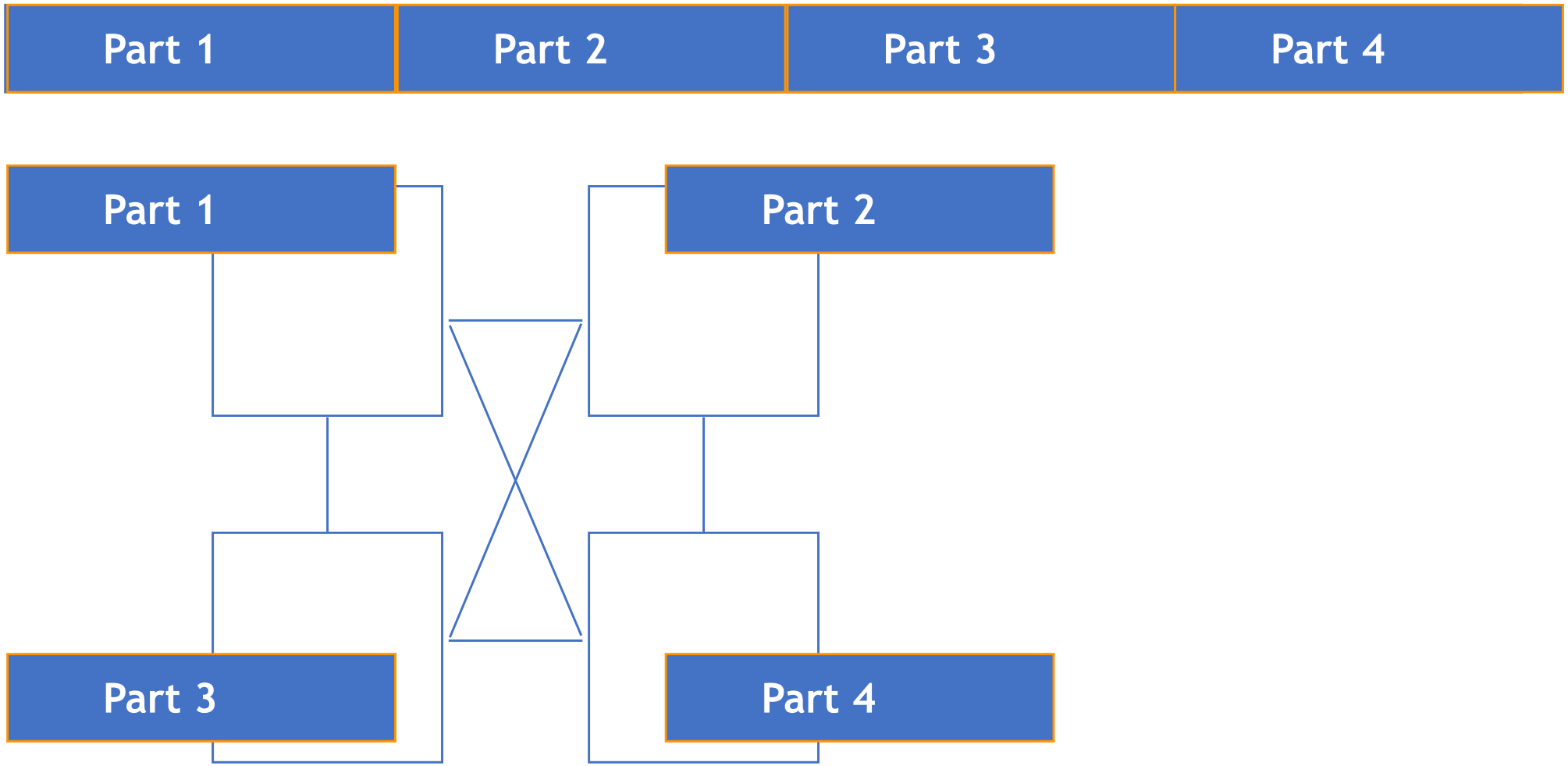
Project 3: DSM EXAMPE Setting

Distributed Shared Memory Abstraction

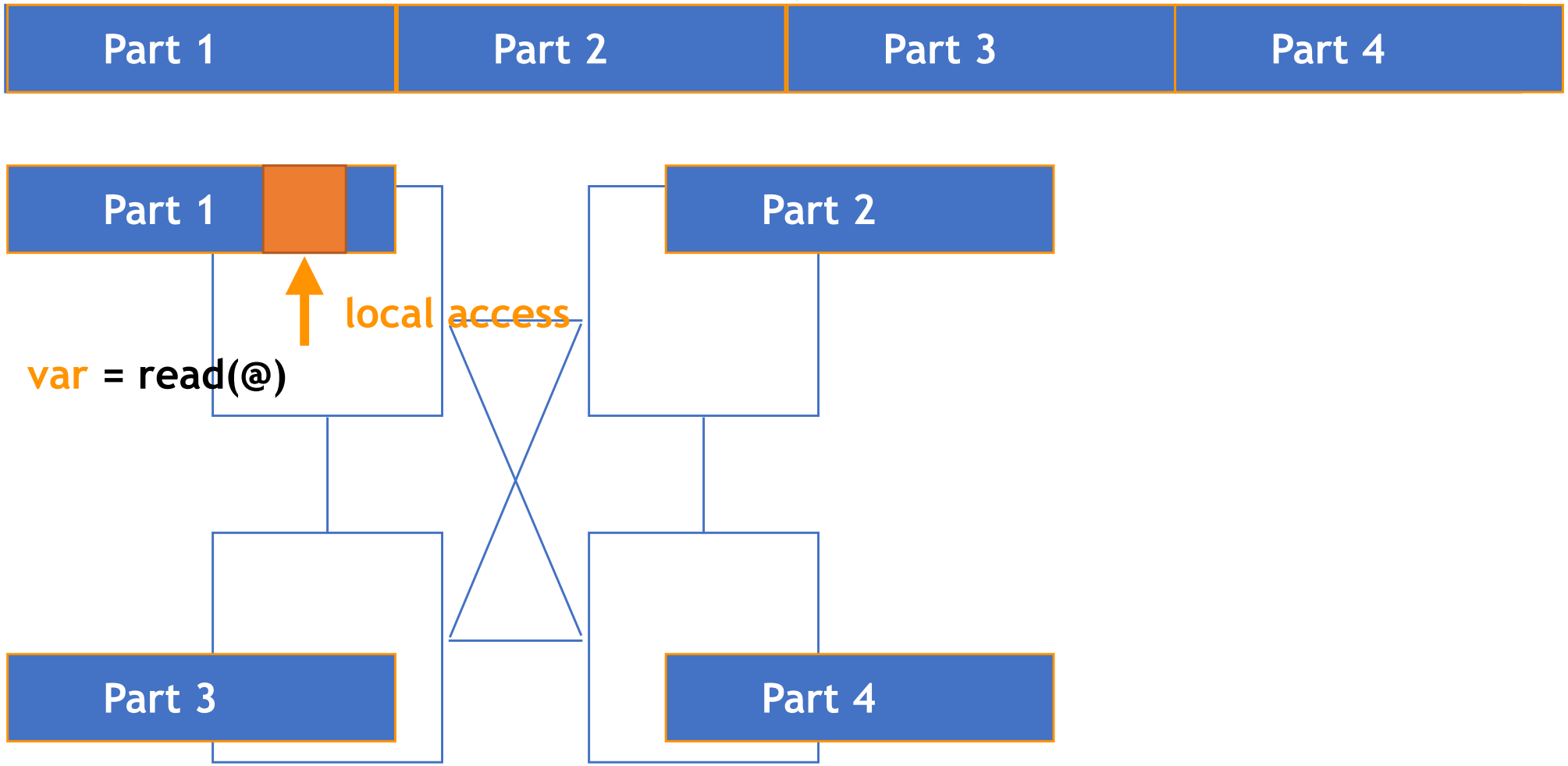


Physical machines in a grid

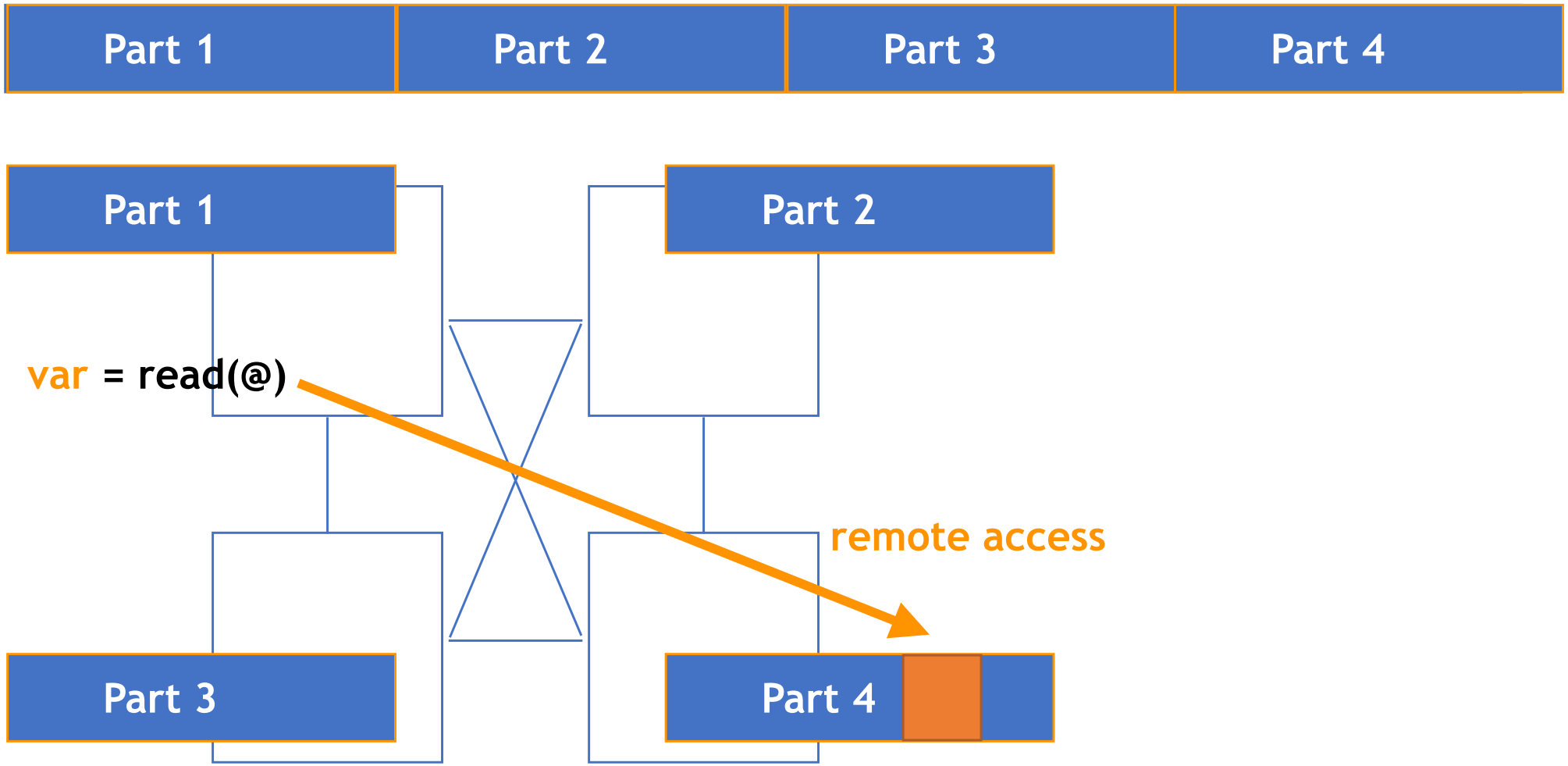
Project 3: DSM EXAMPE with Partitioning



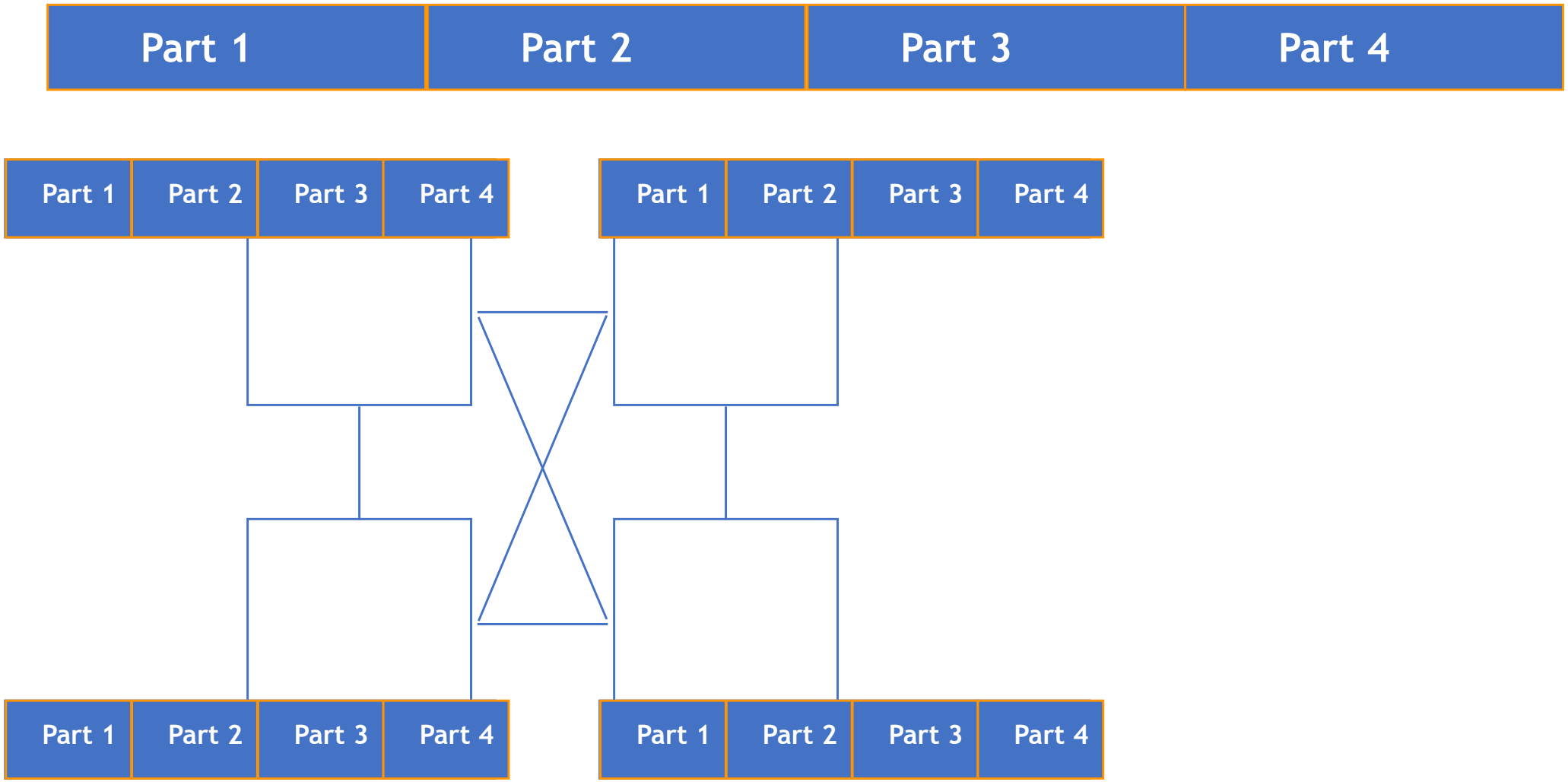
Project 3: DSM EXAMPE with Partitioning



Project 3: DSM EXAMPE with Partitioning



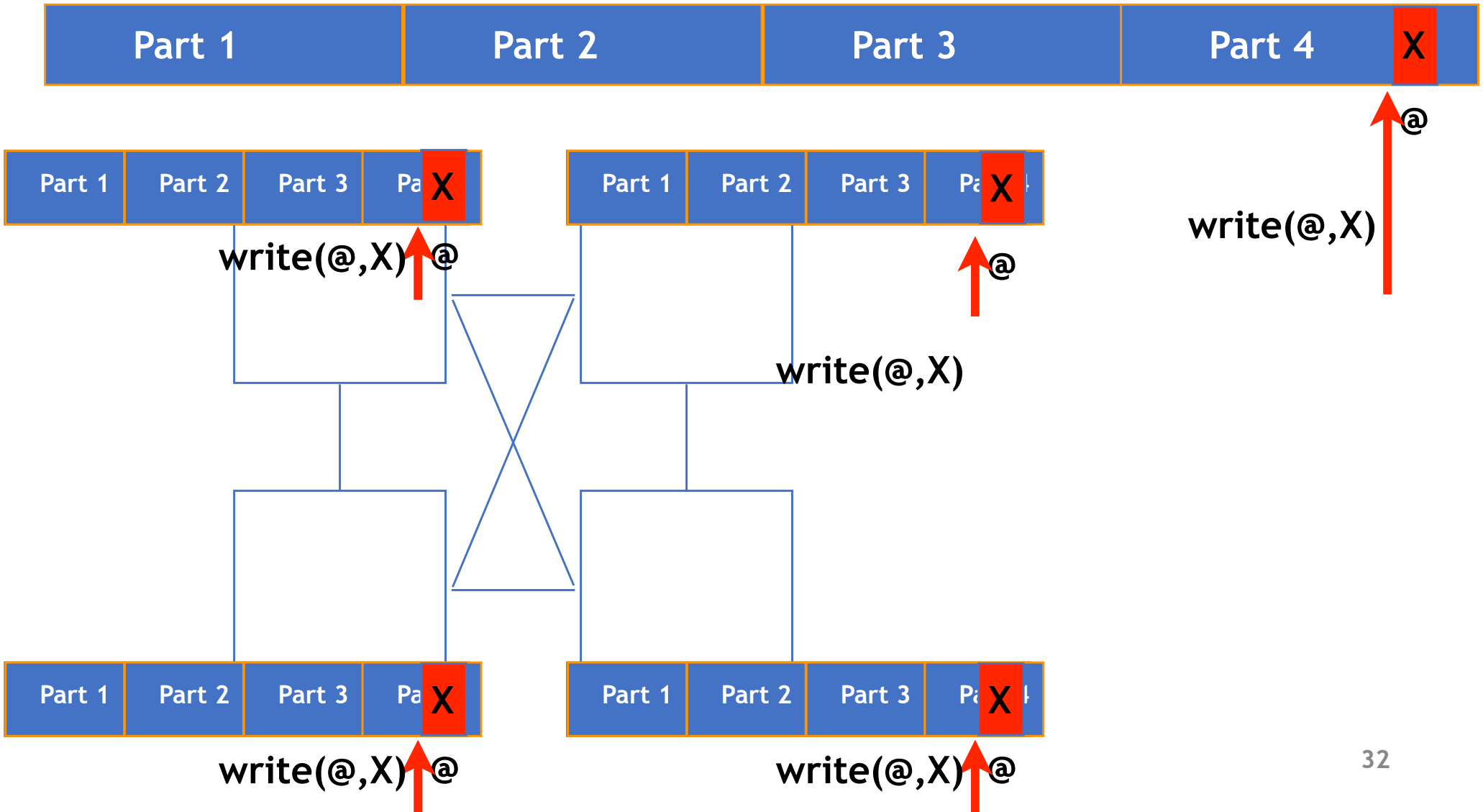
Project 3: DSM EXAMPE with Replication



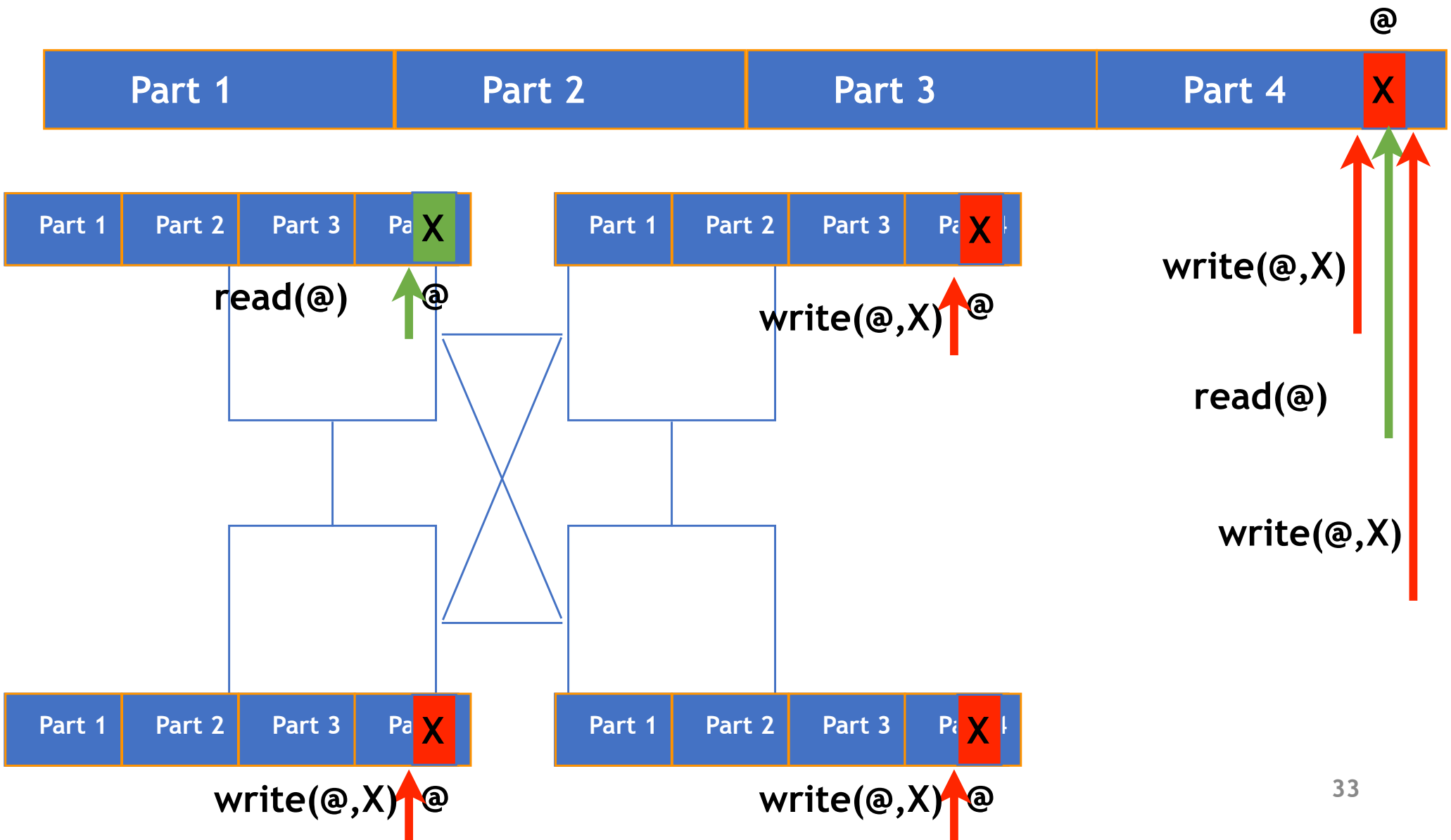
The big problem is consistency
(this is really complicated)

- If there are only read operations, no problem
- The pb is when there are write operations
- What if two write operations concern the same address?
  - Which one will go first?
  - What value will the others observe? (if they read from this address)

Strong consistency



Consistency?



References (on Moodle)

1. Fekete, A. D., Ramamritham, K.: Consistency Models for Replicated Data. In Charron- Bost, B., Pedone, F., Schiper, A., eds. : Replication LNCS 5959. Springer-Verlag, Berlin (2010) 1-17

2. Burckhardt, S.: Principles of Eventual Consistency. Foundations and Trends Programming Languages 1(1-2), 1-150 (2014)

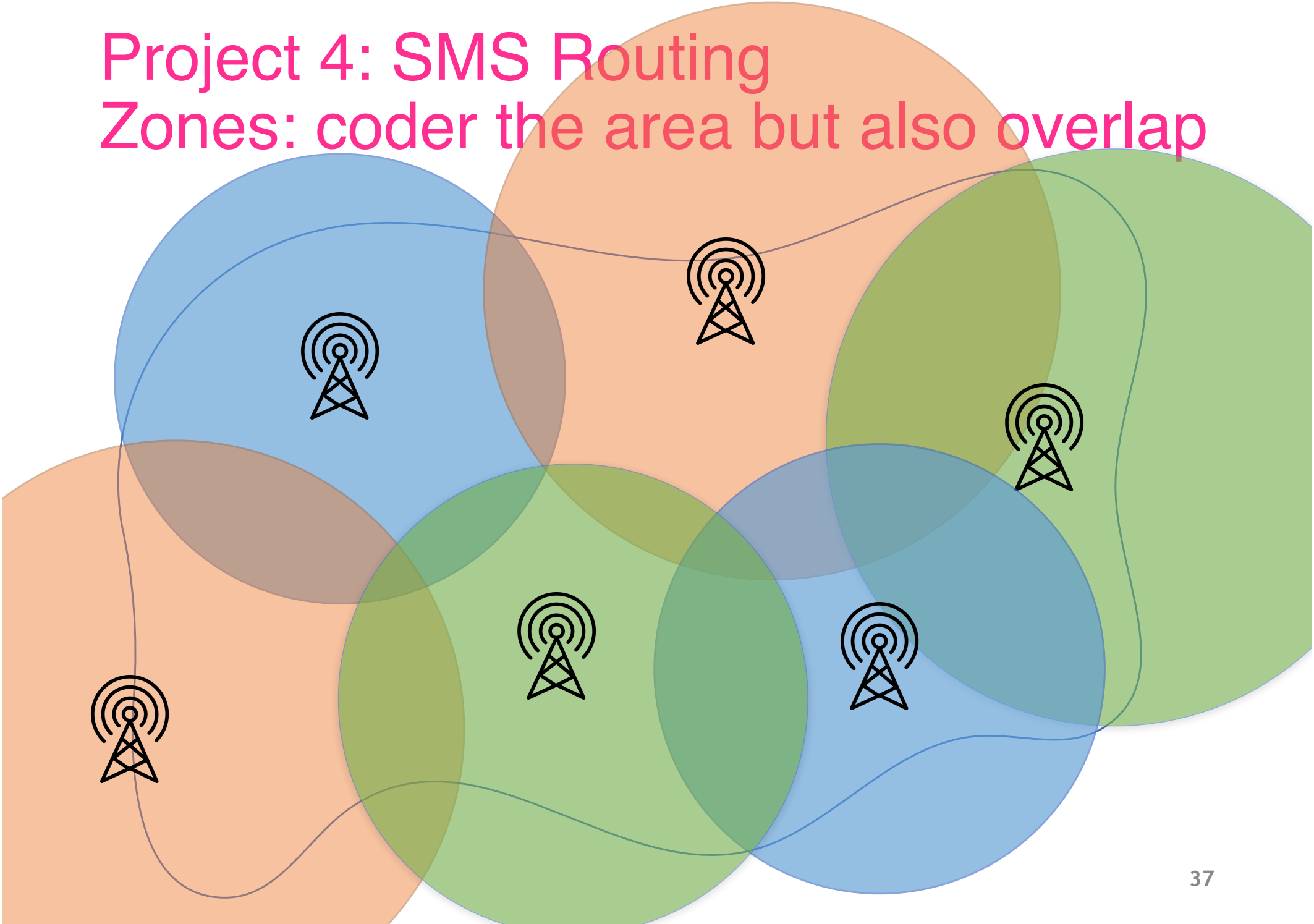
Project 4: SMS Routing

Common setting that you know:

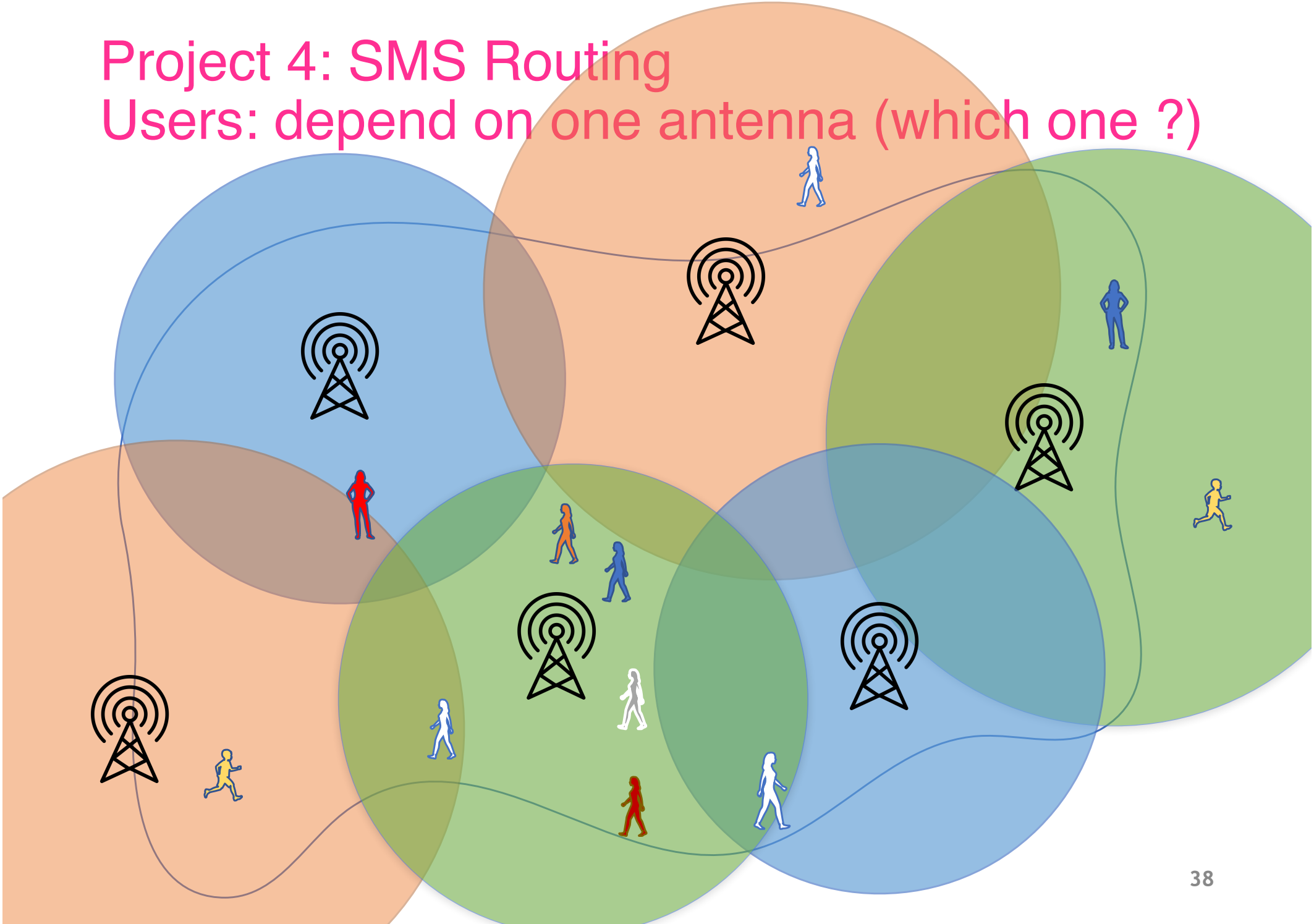
- Geographical area covered with antennas
- Each antenna covers a zone
- Users move around with their phones
- A user sends a message to another user
- The SMS needs to be routed and received

Project 4: SMS Routing Geographical Area

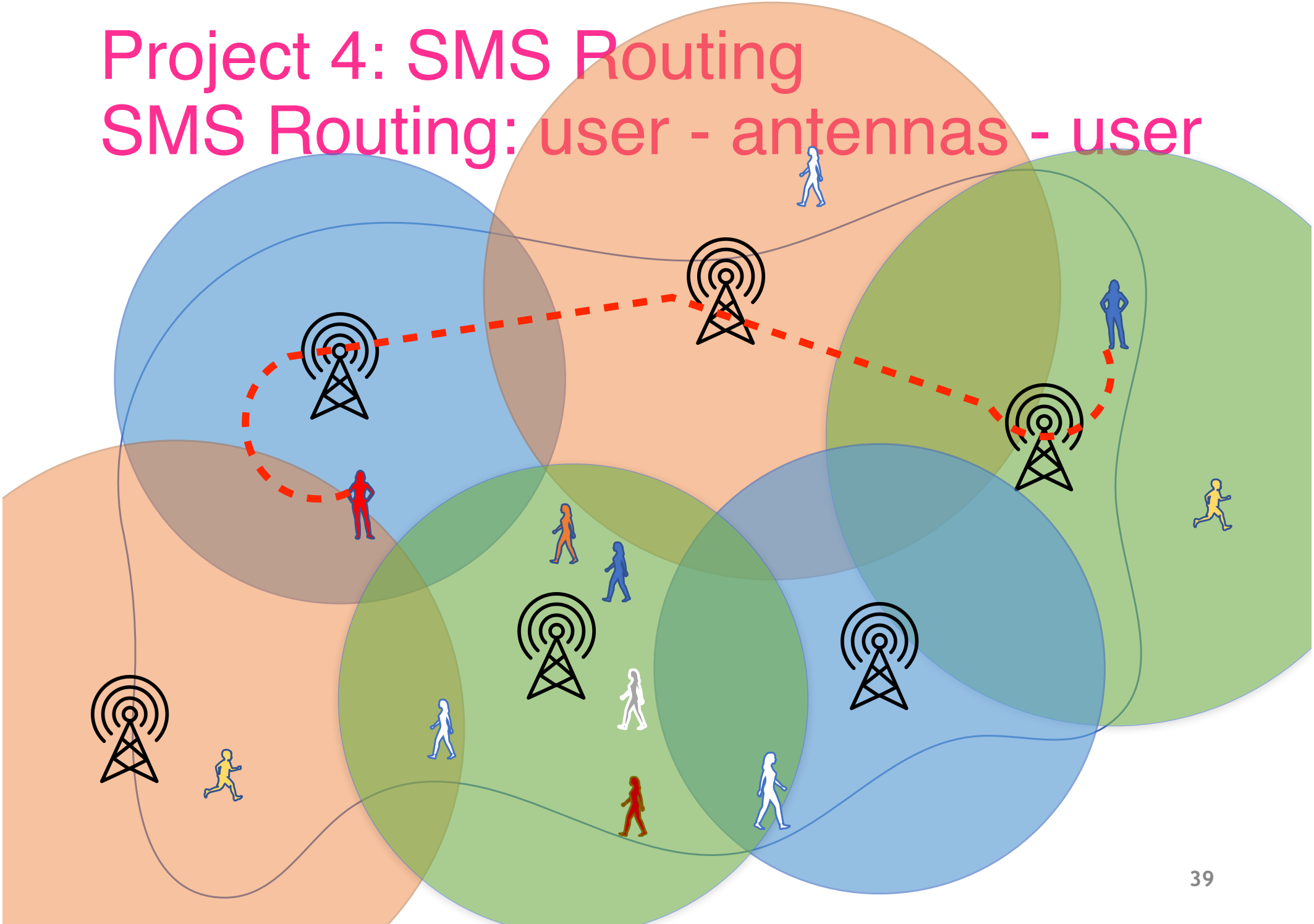
Project 4: SMS Routing
Zones: coder the area but also overlap



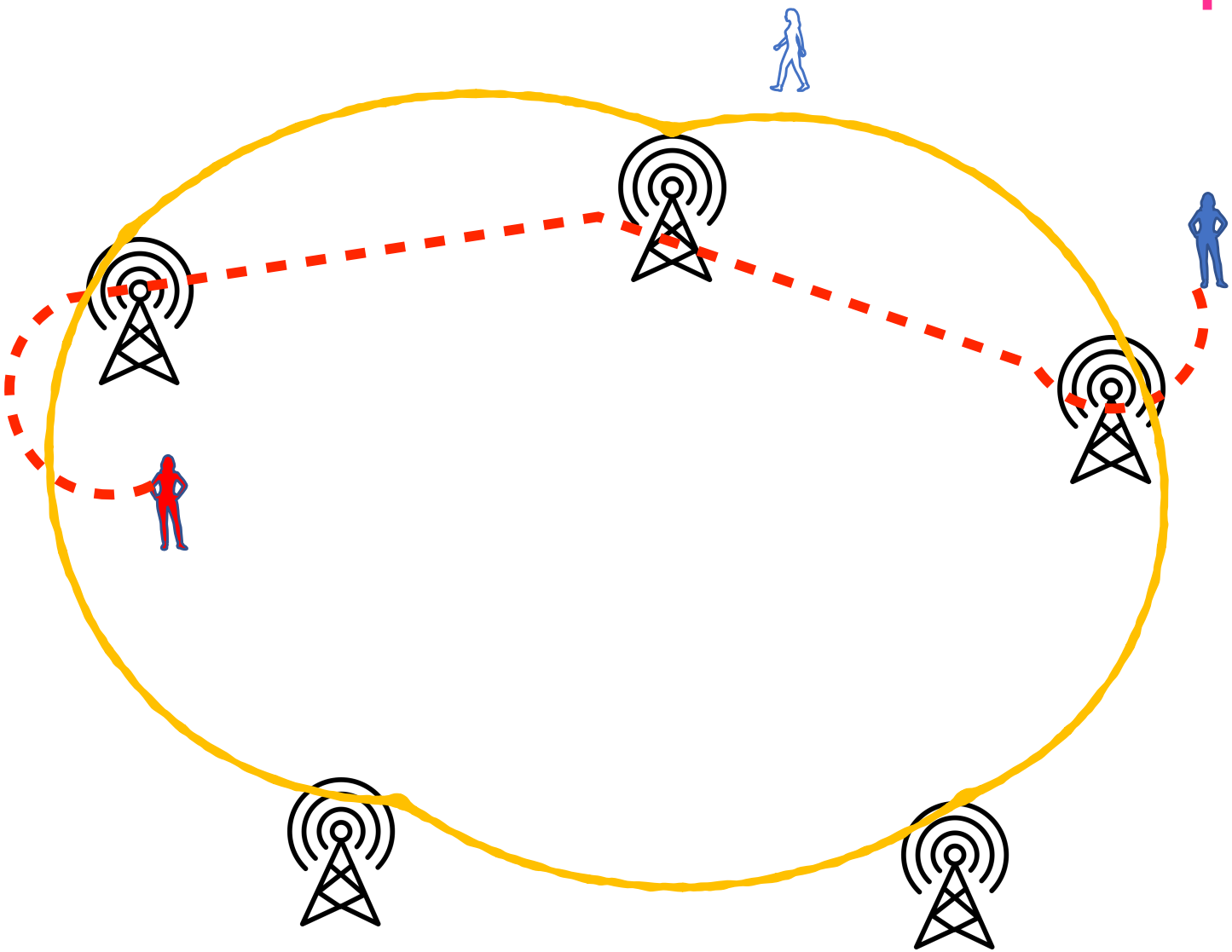
Project 4: SMS Routing
Users: depend on one antenna (which one ?)



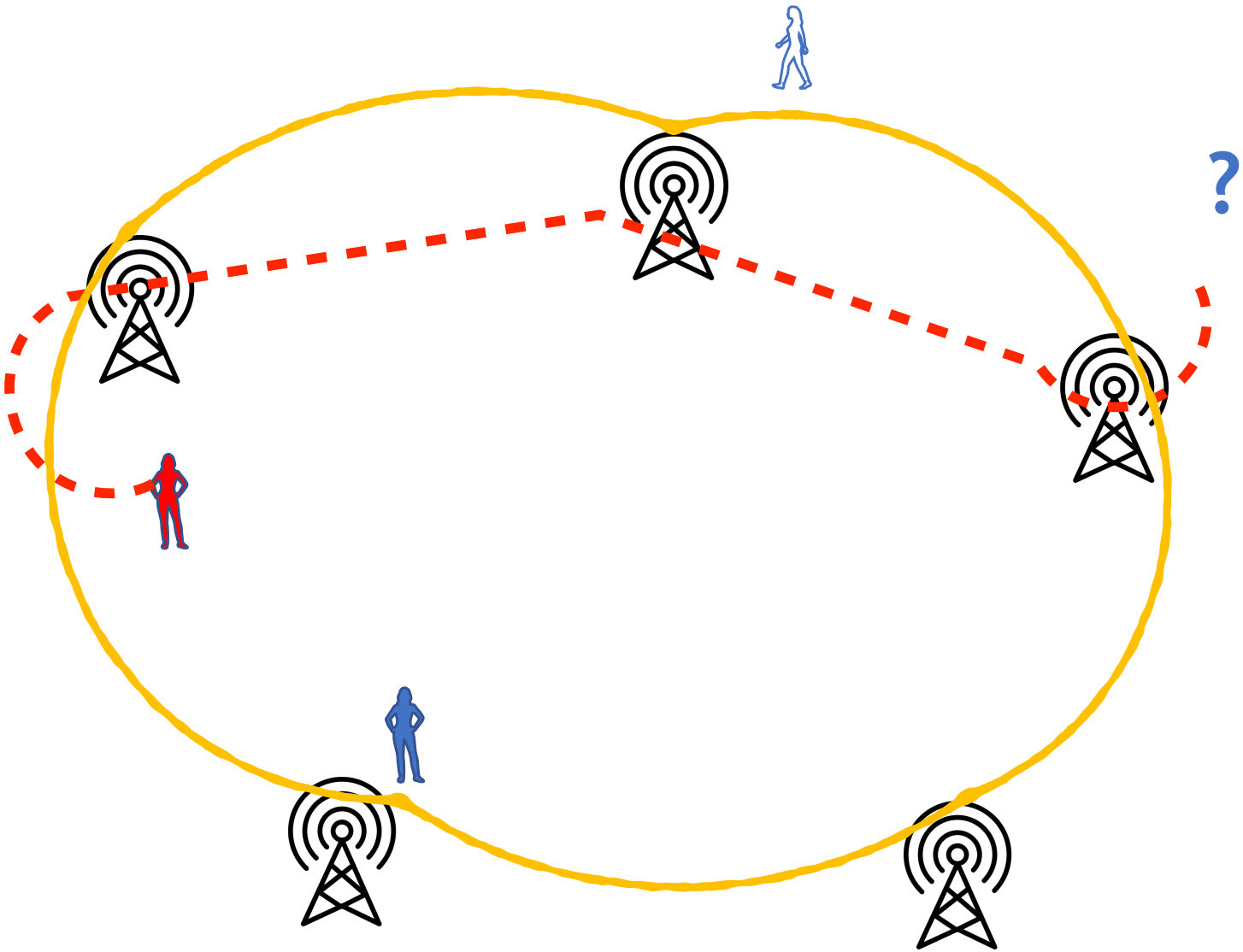
Project 4: SMS Routing
SMS Routing: user - antennas - user



Project 4: SMS Routing
Antenna Interconnexion Example



Project 4: SMS Routing
The user may move... the SMS should follow



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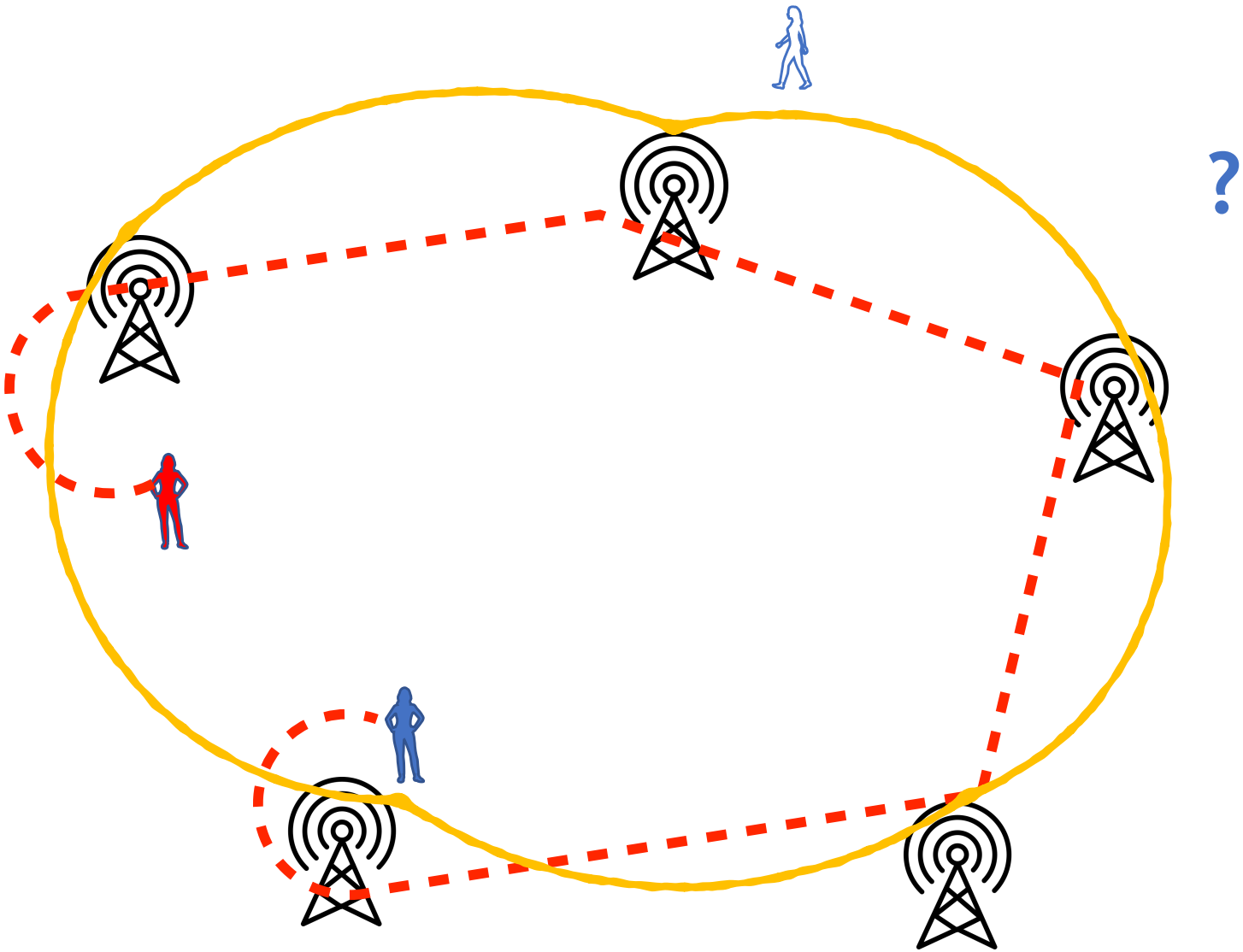

Questions ?